5 Energy is crucial for life and bodily functions, stored in the form of adenosine triphosphate, ATP. Renown as "energy currency", ATP consists of a base, adenine, attached to ribose, to which is attached to a triphosphate group, as shown in Fig. 5.1.

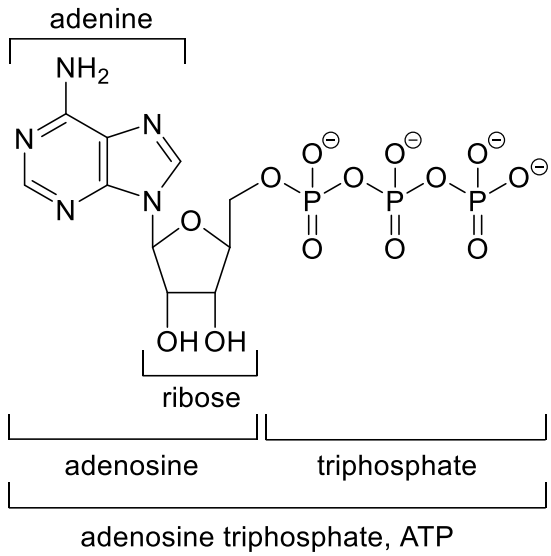

Fig. 5.1

(a) Adenine is made up of a six-membered pyrimidine ring fused with a five-membered imidazole ring, both of which are aromatic and planar.

The imidazole ring specifically contains six π electrons and can exist as an imidazole molecule. Two representations of imidazole molecule are shown in Fig. 5.2.

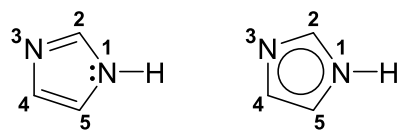

Fig. 5.2

Nitrogen atoms undergo the same type of hybridisation as carbon atoms.

(i) By reference to orbital overlap and the hybridisation of the nitrogen and carbon atoms, suggest how the σ and π bonds are formed in an imidazole molecule.

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..... [3]

(ii) An imidazole molecule is amphoteric as it can function both as an acid and a base.
By considering your answer in **(a)(i)** and its structure or otherwise, suggest and explain:

- The acidic proton is the H atom bonded to N1, not H atoms bonded to carbon atoms.
- N3 acts as the base.

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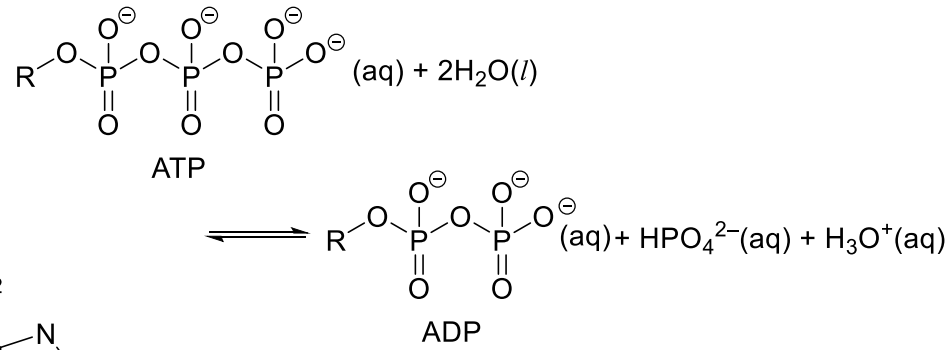
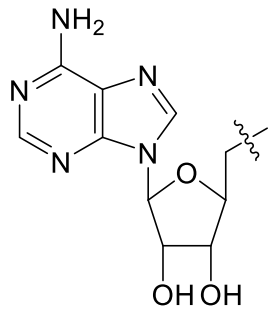
..... [2]

ATP hydrolysis breaks a high-energy P–O bond, releasing energy and forming ADP and HPO_4^{2-}, and is reversible under suitable conditions in reaction 1. The biochemical standard condition (as annotated by the superscript of $^{\ominus\prime}$) applies here where concentrations of all species are defined to be at 1 mol dm^{-3} at pH 7.0, at 37 ºC.

reaction 1



where R =



$\Delta G^{\ominus\prime} = -30.5$ kJ mol^{-1}

$\Delta H^{\ominus\prime} = -20.5$ kJ mol^{-1}

(b) The mechanism of the hydrolysis of ATP proceeds via the following stages:

1. Similar to nucleophilic addition, a nucleophilic attack of water on a phosphorus atom of the terminal phosphate unit while breaking the P=O π bond, forming intermediate 1.
2. The negative charge of the oxygen atom in intermediate 1 is then conferred to reforming of the π-bond in the P=O while another P–O bond involving the phosphorus atom of the terminal phosphate unit is cleaved to form intermediate 2 and ADP.
3. Another water molecule abstracts a proton from intermediate 2.

(i) Suggest why a water molecule is a nucleophile.

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..... [1]

(ii) Complete Fig. 5.3 to suggest the mechanism for this reaction for stages 1 and 2 only. Show the displayed structure of intermediate 2, relevant dipoles, relevant lone pairs of electrons and the movement of electrons by using curly arrows. [3]

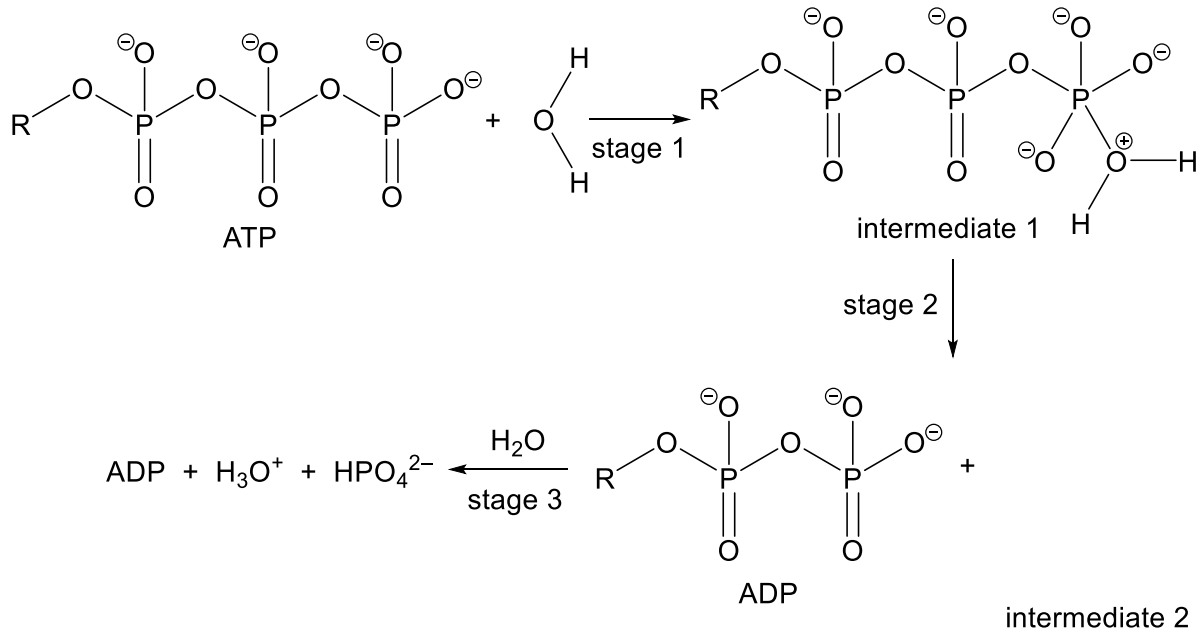

Fig. 5.3

(c) Calculate the entropy change of reaction, $\Delta S^{\ominus\prime}$, at 37 ºC and comment on the sign of $\Delta S^{\ominus\prime}$ obtained.

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..... [2]

(d) Physiological conditions vary depending on the organism, the specific tissue or cell compartment, and the current energy needs for metabolic and other reactions. Table 5.1 shows the concentrations of ATP, ADP and HPO_4^{2-} for various physiological conditions of organism.

Table 5.1

physiological condition of organism	ATP concentration / mol dm^{-3}	ADP concentration / mol dm^{-3}	HPO_4^{2-} concentration / mol dm^{-3}
standard condition	1	1	1
human – resting muscle	8×10^{-3}	9×10^{-6}	4×10^{-3}
human – muscle recovery from severe exercise	8×10^{-3}	7×10^{-6}	1×10^{-3}

(i) Other than concentration to be 1 mol dm^{-3}, state another condition specified by the symbol $^{\ominus}$ when the enthalpy change, entropy change and Gibbs free energy for a reaction are described at 298 K.

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(ii) By considering Table 5.1 or otherwise, suggest why the standard condition is not applicable to most physiological conditions.

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..... [1]

(iii) Equation 1 can be used to compute the actual Gibbs free energy change, ΔG due to ATP hydrolysis under physiological condition.

Equation 1 $\Delta G = \Delta G^{\ominus\prime} + RT \ln Q_c$

where Q_c refers to the reaction quotient which has the same expression as K_c.

Using equation 1, Table 5.1 and the *Data Booklet*, calculate the Gibbs free energy change, ΔG due to ATP hydrolysis in reaction 1 for human muscle of athletes recovering from severe physical exertion at 37 ºC at pH = 7.4.

[2]

In the presence of magnesium ions, the Gibbs free energy change of reaction 1 changes. This is shown in Fig. 5.4.

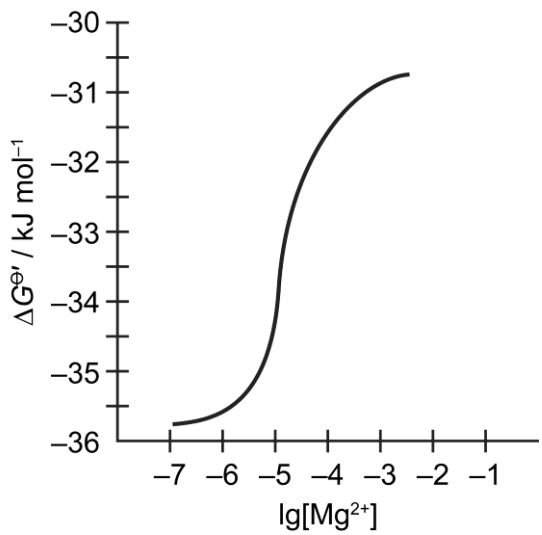

Fig. 5.4

(e) (i) Based on Fig. 5.4, describe how Gibbs free energy change varies with increasing concentration of magnesium ions.

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..... [1]

(ii) ATP can form several complexes with magnesium ions such as the $[MgATP]^{2-}$ complex. The $[MgATP]^{2-}$ complex serves as the substrate for ATPase, the enzyme that catalyses reaction 1.

The rate of reaction 1 using a fixed amount of ATPase is investigated. Experiments are performed using different concentrations of $[MgATP]^{2-}$ complex (substrate) and the rate of each reaction is measured.

Sketch a graph to describe the relationship between the rate of the reaction and substrate concentration, using a fixed amount of ATPase enzyme, in this reaction. Explain your reasoning.

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[Total: 18]